Milestone 02

PDF edition of preserved repository document | docs/milestone_02.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Phase 2 milestone: dataset and benchmark infrastructure

Implemented

General benchmark schemas and backend registry; deterministic configurable generation; identity-aware splits, OOD isolation, checksum/target/leakage validation; three separate data forms; nested budgets; measurable difficulty filters; independent metrics/N*/DER; abstract model interface, failure records, runner, immutable artifacts and four CLI operations.

Tested and verified

54 tests passed on Python 3.12.14, including all 21 original tests. Tests cover generation, seeds/byte reproducibility, JSON schema roundtrips, targets, digit ranges, signed inputs, difficulty bounds, split identities, swapped OOD/operands, trace contamination, target-free views, subsets, metric denominators, missing/malformed/foreign/incomplete trajectories, N*/DER, exception records, pre-call validation, no overwrite and repeat-run predictions.

A non-arithmetic test-only backend verifies generic pipeline reuse.

Development dataset: train 1000; validation 250; IID test 500; OOD test 500.

All targets and leakage checks passed. Nested budgets: 10,25,50,100,250,500,1000.

Split	Examples	Accuracy	Grounded rate	Trajectory validity	Goal success	Artifact
test	500	1.0	1.0	1.0	1.0	results/20261008T005539Z_26822e7d16a44754af415c71c728034d
ood_test	500	1.0	1.0	1.0	1.0	results/20261008T005539Z_d7e71c7dd32e46e5864e15af0d53a552

These are deterministic reference infrastructure results, not learned model results.

No DER or neural data-efficiency evidence exists. Training time is null; no training occurred.

Not implemented

Neural training/downloads, learned baselines, experimental comparisons, interactive experiential learning, confidence intervals, UI, dream engine or published historical scores.

Known limitations

Rejection sampling is unbalanced and exhausts the finite 1x1 family in train; IID evaluation has only 2x1/2x2 examples. Full datasets/traces are in memory. Directory creation prevents overwrite but filesystem-write failures can leave partial directories. Future backends must implement identity/structure/validation/evaluation correctly. Stored schema checks are strict for core traces; benchmark metadata uses backend validation. Python model views are API isolation, not an adversarial filesystem sandbox. model_calls relies on model reporting; timeout records require the model to raise TimeoutError, not a runner-enforced wall-clock timeout. Training budget is metadata/subset identity only: future training must consume that subset. Ruff/MyPy/pytest and hosted CI were not run.

Relevant repository tree

```
configs/experiments/{iid,ood,data_efficiency}.toml
src/mindscape/core/serialization.py
src/mindscape/data/{schemas,backends,generation,splits}.py
src/mindscape/evaluation/{schemas,metrics,evaluator,runner}.py
src/mindscape/models/{base,reference}.py
scripts/{generate_data,dataset,evaluate}.py
datasets/generated/development_v1/{manifest.json,*.jsonl}
results/<unique_id>/{config.json,metrics.json,predictions.jsonl,summary.md}
docs/{benchmark,experimental_protocol,historical_references,milestone_02}.md
tests/{test_foundation,test_benchmark}.py
```

Git

History preserved on main. Phase-2 commits add infrastructure, verification tests, and documented development artifacts. Results remain available locally but are ignored by Git. Final working-tree status and latest commit checked after committing.